PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf)** and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Network meta-analysis for comparative effectiveness of treatments for chronic low back pain disorders: Systematic review protocol
AUTHORS	Belavy, Daniel; Diwan, Ashish; Ford, Jon; Miller, Clint; Hahne, Andrew; Mundell, Niamh; Tagliaferri, Scott; Bowe, Steven; Pedder, Hugo; Saueressig, Tobias; Zhao, Xiaohui; Chen, Xiaolong; Balasundaram, Arun Prasad; Arora, Nitin Kumar; Owen, Patrick

VERSION 1 – REVIEW

REVIEWER	Diogo Pires Escola Superior de Saúde do Alcoitão, Physiotherapy
REVIEW RETURNED	16-Apr-2021

GENERAL COMMENTS	Manuscript ID: bmjopen-2021-051593 The manuscript describes the rationale and methods for developing a systematic review on a current topic of great importance and may have a significant impact on future research and current clinical practice in patients with chronic low back pain. According to the dates reported in the manuscript, the study is ongoing, and therefore my comments seek to clarify some less explicit details and improve the manuscript rather than to suggest major methodological changes. I suggest that the authors consider the following aspects: • Page 4 (Strengths and limitations of this study) - In the first topic please add "chronic low back disorders"; Are there no limitations to reporting? For example, it may be anticipated that there will be difficulties in obtaining full details from the primary studies and communication with the original authors may be complex or impossible. • Page 5, Line 11-12 (Introduction) - I suggest introducing prevalence data at a global level or, if none exist, adding data from other countries and regions beyond Australia. • Page 7, Line 8-10 - I think that including articles in languages whose authors are proficient is not a plausible argument. As it is described, it seems that this was the only criterion, so I suggest that the sentence be rewritten. • Page 7 (Population) - There is some debate in the literature about the definition of some health conditions/syndromes (e.g. radiculopathy or failed back surgery syndrome). Thus, presenting the definitions adopted (supplementary data) may be important. • Page 7 (Interventions and comparators) - I have some doubts about the different types of treatment. I understand the complexity but in some types of defined treatment I think there are overlapping concepts/ definitions. For example, manual traction (included in traction) can be considered a manual therapy. Similarly, it is strange

	to me that McKenzie is considered a distinct type of treatment. Can you clarify whether treatments within each class will be analysed separately? For example, osteopathy will be analysed separately from manual therapy without manipulation? • Page 8 (Outcomes) - Have primary and secondary outcomes been defined? Were all studies that included at least one of the outcomes at some point of follow-up included? Why was health-related quality of life not considered given that it is a core outcome in assessing the effectiveness of interventions in patients with low back pain? • I think there was no restriction on publication dates in the search but that was not clear to me. May you clarify in the manuscript? • Page 9 (search strategy) - Please briefly present the total number of references found through the various methods as outlined in the supplementary material (page 32 and 33) • Page 9 (Data extraction) - I suggest a specific section for study selection and another for data extraction. I think that the procedures and phases of study selection should be presented in more detail. If there are doubts during the selection process about some selection criteria, what will be the procedures? Will the original authors be contacted? Additionally, a flow diagram may help to clarify the process of search and study selection. • Page 9, Line 37-38 (Data extraction) - What type of details about the interventions will be extracted? Please add in parentheses • Page 14 - I would like to suggest a discussion section where the potential impact and relevance of the study for clinical practice, patients and research as well as possible limitations and strengths can be presented.

REVIEWER	Silvia Gianola I.R.C.C.S. Orthopedic Institute Galeazzi, Clinical Epidemiology Unit
REVIEW RETURNED	24-May-2021

GENERAL COMMENTS	Dear authors, Thank you for the opportunity to review this registered protocol of a systematic review with network meta-analysis (NMA). The authors aim to assess the effectiveness of a wide range of common treatments, encompassing pharmacological, non-pharmacological and surgical treatments for chronic low back pain. The PICO is clinically relevant, and I appreciated any effort made to develop such a huge project for treatments in chronic low back pain. However, here I provided my major concerns regarding this NMA protocol: Major concerns 1) The primary concern is the overlap with existing published and ongoing NMA. Please see references below: https://pubmed.ncbi.nlm.nih.gov/32503666/ https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6513465/ https://pubmed.ncbi.nlm.nih.gov/33896808/ https://pubmed.ncbi.nlm.nih.gov/33884409/ https://pubmed.ncbi.nlm.nih.gov/33706636/ https://pubmed.ncbi.nlm.nih.gov/33400430/ https://pubmed.ncbi.nlm.nih.gov/33019465/ https://pubmed.ncbi.nlm.nih.gov/32948543/ https://pubmed.ncbi.nlm.nih.gov/32759244/ https://pubmed.ncbi.nlm.nih.gov/31666220/ https://pubmed.ncbi.nlm.nih.gov/31217314/ https://pubmed.ncbi.nlm.nih.gov/27831982/

	2) Even if the PICO is clinically relevant, the scope of the review seems too broad. Query whether such a broad PIC will produce meaningful conclusions for practice and research. My primary concerns are the specific features of the population and intervention included and consequently, the evaluation of transitivity. These features translate directly to network structure, which has direct influence on effect estimates because inference in NMA incorporates information on trials, and the relationship between those trials (network structure). Here, I am going to deepen my concerns regarding the PIC feature: - Page 7, lines 17-50 > Population: the authors selected a very broad population, including spondylolisthesis, spondylosis, disc herniation, disc degeneration, scoliosis, deformity (e.g. hemivertebrae) and radicular syndromes (e.g. radicular pain [leg pain or sciatica], radiculopathy, spinal stenosis). Query how you could assure transitivity with such heterogenous population. Can results be generalized to all these different populations? For instance, in reference #19, it is stated that "Specific spinal pathology and radicular syndrome are the two distinct LBP triage categories that need to be excluded before a diagnosis of NSLBP, or ordinary backache, can be made for most patients", underlining that these conditions would require different management. Moreover, interventions can be differently provided to different population and so affecting the jointly randomizability in the network. I wonder if results can be clinically transferable when in the same network there are patients with non-specific low back pain and patients with scoliosis or deformity. Shouldn't be assessed separately since treatment can be different? Please see https://pubmed.ncbi.nlm.nih.gov/29573870/ - Pages 7-8, lines 54-24 > Interventions: I am not sure I would consider all of these interventions to be competing options for the clinical decision of what intervention to provide to someone with LBP (and back deformity, which is a different population). Are these nodes realistically compared in practice? Do patients have similar baseline characteristics to be jointly randomizable eg., in both surgery and conservative treatments? Please see https://training.cochrane.org/handbook/current/chapter-11#section-11-3-2-1, where it is stated that "Populations and interventions often need to be considered together given the potential for intransitivity (see Section 11.2.2). A driving principle is that any eligible participant should be eligible for randomization to any included intervention (Salanti 2012, Jansen and Naci 2013)." 3) Looking at Supplemental data A, I have concerns regarding the network construction process: Page 25, lines 10-16> Physical therapy (otherwise not falling into specific treatment combination) (pio) "Into this group fall any interventions that are generic 'physiotherapy' or 'physical therapy' treatments, often at the discretion of the clinician, but otherwise not detailed or defined." I do not understand the clinical meaning of this node. If a treatment does not meet the definition of pre-specified nodes due to unclear

	reporting, I would request additional information to corresponding author, otherwise I would exclude it from the analysis. The risk is to include a node with unclear type of intervention undermining the external validity of your results. Page 20, line 60 > elc_mech: ultrasound therapy and a variety of mechanical therapies: what kind of variety of mechanical therapy? Are ultrasounds similar to vibration therapy - vibration therapy and intermittent pneumatic compression therapy? Page 21, line 52 > Osteopathy – what did the authors consider in this node? Visceral techniques are included? Please, specify. Placing each of these interventions into the same node assumes that the relative effects of these interventions are not importantly different. What evidence did you have for this assumption? My suggestion is to follow recommended guidelines for the network construction process. In addition, upon seeing the network, my concerns are realized because the nodes are broadly defined, apparently also incorporating no information on delivery parameters. Please, add more information. 4) Pages 25-26. Another major concern is regarding the definition of primary and secondary INT components. I'm not really confident that this choice of considering the contribution percentage of an intervention can limit the influence of co-interventions. This means that nodes are built upon the presence of a major contribution by an intervention, but the intervention is anyway a combination of treatments. For example, considering this element of hierarchy for deciding what is the primary intervention: "where a treatment component was mentioned in either the article title or the group subheading was labelled as one of the interventions, then that was taken as the primary intervention component. For example, if the group subheading was called "exercise" but it contained exercise and advice components, then exercise was considered the primary component)." Again "to split true stalemates, the intervention element mentioned first in the treatment description and/or label was taken to be the primary component (e.g. "exercise and advice" = exercise mentioned first and therefore primary component)." Just based your decision on the label wouldn't be misleading and not representing the real intervention? The node might be full of noise and results not representing the real effect of the intervention. How the authors are going to manage, for example, the drug as needed? This is not a primary intervention but participants that assume drugs might have different effects compared to those which do not assume drugs as needed. The authors are going to build the nodes with an unclear clinical rationale. It makes no sense consider exercise as a primary intervention in an arm where, in addition to exercise, the patient receives also advice and drug as needed. Page 26, lines 20-27 "Secondary component of treatments with multiple components: in some cases, a treatment group may have more than two components, but not fall under the multidisciplinary

	definition. In this case, the following principles were followed: only ONE secondary intervention component was included, regardless of how many there were. To qualify as being classified as a secondary treatment component, it needed to represent at least 20% of the total intervention (per judgement of the extractor), otherwise the intervention will be classified as having only a primary intervention with no secondary component." So, are you including more than two treatments? How can we deal with the heterogeneity of treatments in each node? For example, in exercise node are you including RCTs with some assessing only exercise, some exercise and other treatments? If you will do, one does not know what to expect might be compared. 5) Study design: Be careful to include cluster trials and randomized cross-over trials as they can increase some risk of bias and undermine transitivity assumption. Please, better elaborate how will you deal with cluster and crossover designs. For example, cluster trials need adjustments in analysis and the methodological quality requires the assessments of additional items. Please, provide practical insight on the inclusion of these study designs in the model and if the time course Model-Based NMA (MBNMA) take into account the features of these designs. 6) Page 12> Statistical analysis> Assessment of similarity (transitivity). Are you sure you will assess all possible effect modifiers? Consider also baseline psychological conditions, presence of co- interventions, enriched designs, type of low back pain (eg scoliosis, radiculopathy) as reported in reference number #19. How are the authors going to manage different distribution of type of LBP population in specific nodes then others? How will the authors deal with nodes with different population (eg surgery in a scoliosis population versus psychological interventions in non-specific low back pain)? https://training.cochrane.org/handbook/current/chapter-11#section-11-2-2-1 Page 10, lines 10-19 > Data extraction. Be careful imputing medians in order to avoid implausible estimates. According to Cochrane reviews "means and medians can be very different from each other if the data are skewed, and medians are often reported because the data are skewed". Usually "the use of interquartile ranges rather than standard deviations can often be taken as an indicator that the outcomes distribution is skewed" https://handbook-5-1.cochrane.org/chapter_7/7_7_3_5_mediansand_interquartile_ranges.htm Generally, medians and quartiles are often reported when data do not follow a normal distribution https://www.zora.uzh.ch/id/eprint/133040/1/p6_OK.pdf Indeed, the reference number #31 argue that "it still remains a question to which extent it makes sense to use them in a meta-analysis, if the underlying distribution is very asymmetric and one must assume that they don't represent location and dispersion

	adequately” Page 11, line 52> statistical analysis “A minimum of 50 participants will be required per class of treatment for it to be included in meta-analysis”. Is there any clinical or methodological reason? Please add a reference supporting this choice Minor concerns - Data extraction Please, clarify if the same class of studies (ie head to head interventions) will be excluded. - Risk of bias and GRADE Page 11, line 26-33: Please clarify which approach you used to implement the GRADE framework (Georgia Salanti’s group or the GRADE working group? https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7122720/ Please, indicate the operational criteria you will use to form judgements for each domain. Risk of bias: how the authors define the overall judgement? If the GRADE approach using CINEMA was applied, an overall unique judgment for each study would be implemented. - Statistical analysis Page 11, Lines 54-59: Please define how SDs will be imputed. Please, for a clearer readability, describe in another paragraph methods for conventional pair wise meta-analysis. Page 12, from line 56 > Inconsistency. Please describe strategies for dealing in case of source of inconsistency found (eg node splitting, metaregression,…). Please, consider planning also subgroup analysis since pharmacological and non-pharmacological studies as well as surgical and non- surgical may differ in patient and study characteristics and type of biases that may exist. Page 13, from line 22 > Additional assumptions required for analysis of time-course data: this paragraph is not clear for me Page 16 > Network meta-analysis I did not find plans for geometry of the network. Describe methods used to explore the geometry of the treatment network under study and potential biases related to it. This should include how the evidence base has been graphically summarized for presentation, and what characteristics were compiled and used to describe the evidence to readers. It is also missing methods for relative treatment ranking to provide the hierarchy of the best treatments. Please add it.

VERSION 1 – AUTHOR RESPONSE

REVIEWER #1 (Dr. Diogo Pires, Escola Superior de Saúde do Alcoitão, Instituto Politécnico de Setúbal Escola Superior de Saúde)

Reviewer #1: Thank you to the authors for addressing my previous comments. I was satisfied with the responses and changes made. Good job!
Author response: thank-you for your positive comments

REVIEWER #2 (Dr. Silvia Gianola, I.R.C.C.S. Orthopedic Institute Galeazzi, University of Milano-Bicocca)

Reviewer #2: Thank you for the opportunity to review this registered protocol of a systematic review with network meta-analysis (NMA). I have done and discussed this peer review together with Silvia Bargeri and Greta Castellini who are colleagues in our Clinical Epidemiology Unit with expert knowledge on systematic reviews and network meta-analyses to assist and support the final decision. We understand the broad rationale of this work however we still have some concerns that the NMA technique could help to inspect these extensive aims. We identified some major recommendations to be satisfied in order to accept the protocol.
Author response: Thank-you for taking the time to review the protocol with your colleagues. It is in the broader interest of the field to have robust methods published, discussed and scrutinized before a study is finished.

Nonetheless, as we highlight in our letter to the editor, there are often multiple ways to address a problem and, especially in the relatively young NMA field, there are few clear black/white answers to analytical questions. Different research teams will make varying decisions on how to design and implement their study, and this adds to the breadth of literature on particular topics.

RE: "*..to be satisfied in order to accept the protocol*"

With due respect, we point out it is the handling editor and EIC that decide whether a paper is accepted or not, rather than an individual reviewer.

In line with our comments above, whilst we include some changes to our protocol in response to the recommendations of Reviewer 2 below, we rebut a number of comments, and in some cases refer to our prior response where we explained our approach there.

Of note, we will supply our raw data with the published manuscript. This will enable others, such as your group, to conduct re-analysis using alternative methods if they wish. We would welcome a collaboration on such work in the future, it would facilitate mutual understanding and advance the field.

Reviewer #2: Our primary concern is the overlap with the published protocol of Thompson et al.
-Authors declared "Of note, we had registered our systematic review prior to publication of this protocol and it is unclear when their work is due to be published."
However, the PROSPERO protocol registration in Thompson et al. was done on 18 June 2019 (CRD42019138115) whereas the authors submitted their review on PROSPERO on 24th April 2020 with the registration accepted on 5 July 2020 (CRD42020182039).

-In PROSPERO submission authors must identify existing and ongoing reviews to avoid duplication of efforts (Have you searched PROSPERO to identify similar reviews? https://www.crd.york.ac.uk/prospero/#registernew)
Author response: Yes, we checked PROSPERO prior to submitting the review to PROSPERO. Please refer to our prior response for contrasts to the Thompson review.

Beyond this, it is unknown who will publish first: as an example, another research group had a PROSPERO registration for a review on ACL injury management as we were planning our own review on the topic https://www.hs-gesundheit.de/en/research/current-projects/surgery-or-rehabilitation. However, the authors are far from publishing their review and we will submit our finalised review before they do.

As a further point, replication is important in science. We published this NMA (https://bjsm.bmj.com/content/54/21/1279) and a recent publication (https://www.sciencedirect.com/science/article/pii/S1836955321001028) confirmed our original findings, lending strength to our preceding analysis.
With many decisions to be made when implementing a NMA, there is benefit in several approaches being published to test the veracity of the findings and to ensure that the conclusions of a single review are not driven entirely by particular methodological decisions - including those identified by Reviewer 2.

Of note, we will supply our raw data with the published manuscript. This will enable others, such as your group, to conduct re-analysis using alternative methods if they wish. We would welcome a collaboration on such work in the future, it would facilitate mutual understanding and advance the field.

Reviewer #2: Another important concern is about the suitability of the broad PICO that undermine the transitivity assumption necessary to perform the NMA. Method section should be strongly revised.
Author response: With many decisions to be made when implementing a NMA, there is benefit in several approaches being published to test the veracity of the findings and to ensure that the conclusions of a single review are not driven entirely by particular methodological decisions - including those identified by Reviewer 2.

Of note, we will supply our raw data with the published manuscript. This will enable others, such as your group, to conduct re-analysis using alternative methods if they wish. We would welcome a collaboration on such work in the future, it would facilitate mutual understanding and advance the field.

Population:
Reviewer #2: There is heterogeneity in the population included in the eligibility criteria (such as non-specific low back pain and scoliosis). This population is not suitable to receive the same type of interventions (surgery is usually provided for scoliosis but not for non-specific low back pain).
Author response: per the definitions we follow from high-quality outlets (Bardin LD, King P, Maher CG. Diagnostic triage for low back pain: a practical approach for primary care. Med J

Aust 2017;206:268–73. doi:10.5694/mja16.00828), scoliosis at the lumbar spine would fall in the "non-specific" category.

Of note, we focus on the lumbar spine only and the wider issue of scoliosis, which more commonly impacts the thoracic spine (see Weinstein 2008 The Lancet https://doi.org/10.1016/S0140-6736(08)60658-3), will typically not be included.

Reviewer #2: Then, iatrogenic pain such as post-surgical interventions were excluded but "Failed back surgery syndrome" is included even if often this condition is due to surgically related causes (https://www.physio-pedia.com/Failed_Back_Surgery_Syndrome).
Author response: per the text in our document, iatrogenic pain is excluded: "*iatrogenic pain ... will be excluded*"

RE: failed back surgery: Per the reference we provided in our manuscript (Clancy C, Quinn A, Wilson F. The aetiologies of Failed Back Surgery Syndrome: A systematic review. J Back Musculoskelet Rehabil 2017;30:395–402. doi:10.3233/BMR-150318), failed back surgery "is a term applied to a heterogeneous group of individuals with various aetiologies who share a common characteristic of continued pain following one or more spinal operations". Please refer to that publication for more detail.Per the criteria by Bardin et al that we follow, FBSS falls under non-specific back pain.

Interventions:

There is heterogeneity in the node rationale/building. We still have some concerns. Some nodes included heterogeneity of interventions:
Reviewer #2: FANS, paracetamol, opioids, muscle relaxants drugs, anticonvulsants etc..they work with different rationale, including them in the same node undermine the external validity. What about the clinical implications/recommendations?
Author response: Thank you for your comment. As specified in the protocol we will also use a Bayesian class-based model (https://onlinelibrary.wiley.com/doi/book/10.1002/9781118951651) to estimate treatment effects within a class (e.g. Pharmacotherapy (class) and opioids (treatment)).

So we can estimate the “overall effect” of Pharmacotherapy as well as an effect for opioids.

Reviewer #2: manual therapy node included studies investigating osteopathic treatments. However these can include visceral therapies as well as manual therapies (which is the percentage that allow to include visceral component in the manual therapy node? ideally, 100% visceral treatment cannot be included in the node manual therapy)
Author response: Thank you for your comment. Please see our prior response on this point. Further to this, we can conduct a sensitivity analysis excluding osteopathic INTs from the manual therapy node: “*As some osteopathic interventions may include visceral techniques not declared in the original methods of the study, the impact of removing this from the manual therapy node will be examined.*”

Reviewer #2: multidisciplinary node (exercise, stress management, relaxation and imagery, meditation and aspects of psychological therapies, medication management, family member participation implemented as one package of treatment) overlap with other nodes such as multicomponent node (e.g., exercise that could be the primary component with a secondary component, example exercise 80% with stress management 20%)
Author response: Thank you for your comment. Multidisciplinary pain management is a well defined intervention, per the references and definitions we supply in Supplemental Data A.

Here we insert the excerpt from our definitions in Supplemental Data A:

“*If a study labelled its intervention as multidisciplinary pain management, then this was considered multidisciplinary pain management. Other studies may have combined individual interventions into a multidisciplinary program but did not specifically label it as multidisciplinary pain management. In this case, if the reviewers agreed that the intervention included a minimum of education, exercise, psychological therapies delivered by a multidisciplinary clinician team (at least 2 clinicians from different fields), this was classified as ‘multidisciplinary pain management’. Otherwise, these interventions were classified under 'treatment combinations'*.”

In our experience thus far in classifying INTs, we did not have major challenges separating multidisciplinary pain management from INTs that have multiple components.

Reviewer #2: unclear definition and ambiguity of nodes/mixed interventions. For example, which is the difference between multidisciplinary node and a node with primary and secondary component? Then, which is the difference between Physical therapy and multidisciplinary node? Nodes components are arbitrary chosen and can be subjective to a posteriori changes driven by evidence (primary and secondary components): personnel subjective judgment are based on what? time for dose, sessions, scheduling or duration of whole intervention?
Author response: Please see the detailed information we provide in the Supplemental Data A, especially the section "Definition of primary and secondary INT components". That section is our *a priori* definition, agreed amongst the team in the early stages of the project, of how we approach the independent assessment by at least two skilled reviewers on how the INTs were classified.

For your ease of reference, we paste here the definitions of multidisciplinary versus physical therapy

Multidisciplinary (multidisciplinary pain management) (mul)

Multidisciplinary pain management incorporates a number of intervention types, such as education (e.g., mechanisms of chronic pain, anatomy), goal setting, exercise, stress management, relaxation and imagery, meditation and aspects of psychological therapies, medication management, family member participation implemented as one package of treatment [104–106]. These may be done as individual sessions or as group sessions. If a study labelled its intervention as multidisciplinary pain management, then this was considered multidisciplinary

pain management. Other studies may have combined individual interventions into a multidisciplinary program but did not specifically label it as multidisciplinary pain management. In this case, if the reviewers agreed that the intervention included a minimum of education, exercise, psychological therapies delivered by a multidisciplinary clinician team (at least 2 clinicians from different fields), this was classified as 'multidisciplinary pain management'. Otherwise, these interventions were classified under 'treatment combinations' (below).

Physical therapy (otherwise not falling into specific treatment combination) (pio)

Into this group fall any interventions that are generic 'physiotherapy' or 'physical therapy' treatments, often at the discretion of the clinician, but otherwise not detailed or defined.

We argue this is a well explained, robust, transparent and detailed approach to handling the INTs.

Reviewer #2: the percentage of threshold used (20%?) to choose the primary or secondary components don't reflect the reporting of primary studies. Few or none of the RCTs declare the percentage of performing an intervention. This undermine the internal and external validity of treatment nodes.
Author response: In our work so far, the expert reviewers/extractors have been able to apply our criteria well with discussion amongst the wider team as needed for conflicts / differing viewpoints.

We believe our a-priori defined approach is transparent, well-justified and clear. Yes, some authors may take a different approach, and they are welcome to do so.

Based on the above considerations we recommend:
Reviewer #2: In the main analysis to perform the subgroup analysis (by population) analysis, not only in case of inconsistency; Please, add this in the methods.
Author response: We responded to this in our prior response. This is handled as part of pre-planned sub-group analyses: "*we will perform subgroup analyses ... by type of low back disorder (e.g., non-specific chronic low back pain, radicular syndrome)*"

Reviewer #2: in the main analysis to perform the sensitivity analysis without the multicomponent interventions (i.e., arms with 80% of manual therapy and 20% of exercise), not only in case of inconsistency (or cited as limitation in the discussion section) in order to run only single interventions (e.g., 100% manual therapy); Please, add this in the methods.
Author response: please see our response above regarding a prior paper which in our view incorrectly excluded many studies in their main analysis, potentially leading to biased effects estimates.

We believe this is best handled as a sensitivity analysis of the wider dataset.

We will add a sensitivity analysis on this point: "*Secondary treatment components (see Supplemental Data A): the impact on effect estimates of when secondary treatments are included will be assessed via a sensitivity analysis excluding those interventions with a secondary treatment component.*"

Thank-you for the suggestion.

Reviewer #2: FANS, paracetamol, opioids, muscle relaxants drugs, anticonvulsants etc.. should be presented in separate nodes;
Author response: please see our response above regarding the publication of our dataset and our willingness to collaborate with researchers who are interested in modelling the data differently.

There may be different equally justified construction of INT nodes.

Reviewer #2: be careful in the assessment of imputation of medians. “If a median is available occasionally can be used directly in meta-analyses…Wan and colleagues proposed a formula for imputing a missing mean value based on the lower quartile, median and upper quartile summary statistics (Wan et al 2014) ..assuming normally distributed outcomes” https://training.cochrane.org/handbook/current/chapter-06#section-6-5-2-9 How did the authors plan to verify the normal distribution? In order to be more conservative we strongly recommend to avoid the use of medians if normal distribution cannot be proved, this is more true when authors expect to include over 700 trials. In fact, in our recent NMA on acute and subacute low back pain we found that the use of medians or unusable outcome data was present in around 40% trials https://pubmed.ncbi.nlm.nih.gov/33849907. Imputation of one/two studies in a pairwise meta-analysis can be plausible (occasionally) but what about the imputation of medians in 40% (more or less) of 700 studies?
Author response: In our current extracted data set we found that only around 3% of the studies were imputed due to a median. As the Cochrane collaboration writes: ”*However, imputation may be reasonable for a small proportion of studies comprising a small proportion of the data if it enables them to be combined with other studies for which full data are available. Sensitivity analyses should be used to assess the impact of changing the assumptions made.*” (https://training.cochrane.org/handbook/current/chapter-06#section-6-5-2-7) Hence we will perform a sensitivity analysis by excluding studies with imputed values as stated in our protocol paper.

This was included in the sensitivity analyses in the previously revised submission in the line "• *Excluding studies with imputed missing standard deviations and imputed medians*."

VERSION 2 – REVIEW

REVIEWER	Diogo Pires Escola Superior de Saúde do Alcoitão, Physiotherapy
REVIEW RETURNED	20-Sep-2021

GENERAL COMMENTS	Thank you to the authors for addressing my previous comments. I was satisfied with the responses and changes made. Good job!

REVIEWER	Silvia Gianola I.R.C.C.S. Orthopedic Institute Galeazzi, Clinical Epidemiology Unit
REVIEW RETURNED	30-Sep-2021

GENERAL COMMENTS	Thank you for the opportunity to review this registered protocol of a systematic review with network meta-analysis (NMA). I have done and discussed this peer review together with Silvia Bargeri and Greta Castellini who are colleagues in our Clinical Epidemiology Unit with expert knowledge on systematic reviews and network meta-analyses to assist and support the final decision. We understand the broad rationale of this work however we still have some concerns that the NMA technique could help to inspect these extensive aims. We identified some major recommendations to be satisfied in order to accept the protocol. For authors 1) Our primary concern is the overlap with the published protocol of Thompson et al. -Authors declared “Of note, we had registered our systematic review prior to publication of this protocol and it is it is unclear when their

	work is due to be published." However, the PROSPERO protocol registration in Thompson et al. was done on 18 June 2019 (CRD42019138115) whereas the authors submitted their review on PROSPERO on 24th April 2020 with the registration accepted on 5 July 2020 (CRD42020182039). -In PROSPERO submission authors must identify existing and ongoing reviews to avoid duplication of efforts (Have you searched PROSPERO to identify similar reviews? https://www.crd.york.ac.uk/prospero/#registernew) 2) Another important concern is about the suitability of the broad PICO that undermine the transitivity assumption necessary to perform the NMA. Method section should be strongly revised. Population: 1-There is heterogeneity in the population included in the eligibility criteria (such as non-specific low back pain and scoliosis). This population is not suitable to receive the same type of interventions (surgery is usually provided for scoliosis but not for non-specific low back pain). 2-Then, iatrogenic pain such as post-surgical interventions were excluded but "Failed back surgery syndrome" is included even if often this condition is due to surgically related causes (https://www.physio-pedia.com/Failed_Back_Surgery_Syndrome). Interventions: There is heterogeneity in the node rationale/building. We still have some concerns: Some nodes included heterogeneity of interventions: • FANS, paracetamol, opioids, muscle relaxants drugs, anticonvulsants etc..they work with different rationale, including them in the same node undermine the external validity. What about the clinical implications/recommendations? • manual therapy node included studies investigating osteopathic treatments. However these can include visceral therapies as well as manual therapies (which is the percentage that allow to include visceral component in the manual therapy node? ideally, 100% visceral treatment cannot be included in the node manual therapy) • multidisciplinary node (exercise, stress management, relaxation and imagery, meditation and aspects of psychological therapies, medication management, family member participation implemented as one package of treatment) overlap with other nodes such as multicomponent node (e.g., exercise that could be the primary component with a secondary component, example exercise 80% with stress management 20%) • unclear definition and ambiguity of nodes/mixed interventions. For example, which is the difference between multidisciplinary node and a node with primary and secondary component? Then, which is the difference between Physical therapy and multidisciplinary node? Nodes components are arbitrary chosen and can be subjective to a posteriori changes driven by evidence (primary and secondary components): • personnel subjective judgment are based on what? time for dose, sessions, scheduling or duration of whole intervention? • the percentage of threshold used (20%?) to choose the primary or secondary components don't reflect the reporting of primary studies. Few or none of the RCTs declare the percentage of performing an intervention. This undermine the internal and external validity of

	treatment nodes. Based on the above considerations we recommend: 1- in the main analysis to perform the subgroup analysis (by population) analysis, not only in case of inconsistency; Please, add this in the methods. 2- in the main analysis to perform the sensitivity analysis without the multicomponent interventions (i.e., arms with 80% of manual therapy and 20% of exercise), not only in case of inconsistency (or cited as limitation in the discussion section) in order to run only single interventions (e.g., 100% manual therapy); Please, add this in the methods. 3- FANS, paracetamol, opioids, muscle relaxants drugs, anticonvulsants etc.. should be presented in separate nodes; 4- be careful in the assessment of imputation of medians. "If a median is available occasionally can be used directly in meta-analyses…Wan and colleagues proposed a formula for imputing a missing mean value based on the lower quartile, median and upper quartile summary statistics (Wan et al 2014) ..assuming normally distributed outcomes" https://training.cochrane.org/handbook/current/chapter-06#section-6-5-2-9 How did the authors plan to verify the normal distribution? In order to be more conservative we strongly recommend to avoid the use of medians if normal distribution cannot be proved, this is more true when authors expect to include over 700 trials. In fact, in our recent NMA on acute and subacute low back pain we found that the use of medians or unusable outcome data was present in around 40% trials https://pubmed.ncbi.nlm.nih.gov/33849907. Imputation of one/two studies in a pairwise meta-analysis can be plausible (occasionally) but what about the imputation of medians in 40% (more or less) of 700 studies?

VERSION 2 – AUTHOR RESPONSE

REVIEWER #1 (Dr. Diogo Pires, Escola Superior de Saúde do Alcoitão, Instituto Politécnico de Setúbal Escola Superior de Saúde)

Reviewer #1: Thank you to the authors for addressing my previous comments. I was satisfied with the responses and changes made. Good job!

Author response: thank-you for your positive comments

REVIEWER #2 (Dr. Silvia Gianola, I.R.C.C.S. Orthopedic Institute Galeazzi, University of Milano-Bicocca)

Reviewer #2: Thank you for the opportunity to review this registered protocol of a systematic review with network meta-analysis (NMA). I have done and discussed this peer review together with Silvia Bargeri and Greta Castellini who are colleagues in our Clinical Epidemiology Unit with expert knowledge on systematic reviews and network meta-analyses to assist and support the final decision. We understand the broad rationale of this work however we still have some concerns that the NMA

technique could help to inspect these extensive aims. We identified some major recommendations to be satisfied in order to accept the protocol.

Author response: Thank-you for taking the time to review the protocol with your colleagues. It is in the broader interest of the field to have robust methods published, discussed and scrutinized before a study is finished.

Nonetheless, as we highlight in our letter to the editor, there are often multiple ways to address a problem and, especially in the relatively young NMA field, there are few clear black/white answers to analytical questions. Different research teams will make varying decisions on how to design and implement their study, and this adds to the breadth of literature on particular topics.

RE: "..to be satisfied in order to accept the protocol"

With due respect, we point out it is the handling editor and EIC that decide whether a paper is accepted or not, rather than an individual reviewer.

In line with our comments above, whilst we include some changes to our protocol in response to the recommendations of Reviewer 2 below, we rebut a number of comments, and in some cases refer to our prior response where we explained our approach there.

Of note, we will supply our raw data with the published manuscript. This will enable others, such as your group, to conduct re-analysis using alternative methods if they wish. We would welcome a collaboration on such work in the future, it would facilitate mutual understanding and advance the field.

Reviewer #2: Our primary concern is the overlap with the published protocol of Thompson et al.

-Authors declared "Of note, we had registered our systematic review prior to publication of this protocol and it is unclear when their work is due to be published."

However, the PROSPERO protocol registration in Thompson et al. was done on 18 June 2019 (CRD42019138115) whereas the authors submitted their review on PROSPERO on 24th April 2020 with the registration accepted on 5 July 2020 (CRD42020182039).

-In PROSPERO submission authors must identify existing and ongoing reviews to avoid duplication of efforts (Have you searched PROSPERO to identify similar reviews? https://www.crd.york.ac.uk/prospero/#registernew)

Author response: Yes, we checked PROSPERO prior to submitting the review to PROSPERO. Please refer to our prior response for contrasts to the Thompson review.

Beyond this, it is unknown who will publish first: as an example, another research group had a PROSPERO registration for a review on ACL injury management as we were planning our own review on the topic https://www.hs-gesundheit.de/en/research/current-projects/surgery-or-

rehabilitation. However, the authors are far from publishing their review and we will submit our finalised review before they do.

As a further point, replication is important in science. We published this NMA (https://bjsm.bmj.com/content/54/21/1279) and a recent publication (https://www.sciencedirect.com/science/article/pii/S1836955321001028) confirmed our original findings, lending strength to our preceding analysis.

With many decisions to be made when implementing a NMA, there is benefit in several approaches being published to test the veracity of the findings and to ensure that the conclusions of a single review are not driven entirely by particular methodological decisions - including those identified by Reviewer 2.

Of note, we will supply our raw data with the published manuscript. This will enable others, such as your group, to conduct re-analysis using alternative methods if they wish. We would welcome a collaboration on such work in the future, it would facilitate mutual understanding and advance the field.

Reviewer #2: Another important concern is about the suitability of the broad PICO that undermine the transitivity assumption necessary to perform the NMA. Method section should be strongly revised.

Author response: With many decisions to be made when implementing a NMA, there is benefit in several approaches being published to test the veracity of the findings and to ensure that the conclusions of a single review are not driven entirely by particular methodological decisions - including those identified by Reviewer 2.

Of note, we will supply our raw data with the published manuscript. This will enable others, such as your group, to conduct re-analysis using alternative methods if they wish. We would welcome a collaboration on such work in the future, it would facilitate mutual understanding and advance the field.

Population:

Reviewer #2: There is heterogeneity in the population included in the eligibility criteria (such as non-specific low back pain and scoliosis). This population is not suitable to receive the same type of interventions (surgery is usually provided for scoliosis but not for non-specific low back pain).

Author response: per the definitions we follow from high-quality outlets (Bardin LD, King P, Maher CG. Diagnostic triage for low back pain: a practical approach for primary care. Med J Aust 2017;206:268–73. doi:10.5694/mja16.00828), scoliosis at the lumbar spine would fall in the “non-specific” category.

Of note, we focus on the lumbar spine only and the wider issue of scoliosis, which more commonly impacts the thoracic spine (see Weinstein 2008 The Lancet https://doi.org/10.1016/S0140-6736(08)60658-3), will typically not be included.

Reviewer #2: Then, iatrogenic pain such as post-surgical interventions were excluded but “Failed back surgery syndrome” is included even if often this condition is due to surgically related causes (https://www.physio-pedia.com/Failed_Back_Surgery_Syndrome).

Author response: per the text in our document, iatrogenic pain is excluded: "iatrogenic pain ... will be excluded"

RE: failed back surgery: Per the reference we provided in our manuscript (Clancy C, Quinn A, Wilson F. The aetiologies of Failed Back Surgery Syndrome: A systematic review. J Back Musculoskelet Rehabil 2017;30:395–402. doi:10.3233/BMR-150318), failed back surgery "is a term applied to a heterogeneous group of individuals with various aetiologies who share a common characteristic of continued pain following one or more spinal operations". Please refer to that publication for more detail.Per the criteria by Bardin et al that we follow, FBSS falls under non-specific back pain.

Interventions:

There is heterogeneity in the node rationale/building. We still have some concerns. Some nodes included heterogeneity of interventions:

Reviewer #2: FANS, paracetamol, opioids, muscle relaxants drugs, anticonvulsants etc..they work with different rationale, including them in the same node undermine the external validity. What about the clinical implications/recommendations?

Author response: Thank you for your comment. As specified in the protocol we will also use a Bayesian class-based model (https://onlinelibrary.wiley.com/doi/book/10.1002/9781118951651) to estimate treatment effects within a class (e.g. Pharmacotherapy (class) and opioids (treatment)).

So we can estimate the “overall effect” of Pharmacotherapy as well as an effect for opioids.

Reviewer #2: manual therapy node included studies investigating osteopathic treatments. However these can include visceral therapies as well as manual therapies (which is the percentage that allow to include visceral component in the manual therapy node? ideally, 100% visceral treatment cannot be included in the node manual therapy)

Author response: Thank you for your comment. Please see our prior response on this point. Further to this, we can conduct a sensitivity analysis excluding osteopathic INTs from the manual therapy node: “As some osteopathic interventions may include visceral techniques not declared in the original methods of the study, the impact of removing this from the manual therapy node will be examined.”

Reviewer #2: multidisciplinary node (exercise, stress management, relaxation and imagery, meditation and aspects of psychological therapies, medication management, family member

participation implemented as one package of treatment) overlap with other nodes such as multicomponent node (e.g., exercise that could be the primary component with a secondary component, example exercise 80% with stress management 20%)

Author response: Thank you for your comment. Multidisciplinary pain management is a well defined intervention, per the references and definitions we supply in Supplemental Data A.

Here we insert the excerpt from our definitions in Supplemental Data A:

“If a study labelled its intervention as multidisciplinary pain management, then this was considered multidisciplinary pain management. Other studies may have combined individual interventions into a multidisciplinary program but did not specifically label it as multidisciplinary pain management. In this case, if the reviewers agreed that the intervention included a minimum of education, exercise, psychological therapies delivered by a multidisciplinary clinician team (at least 2 clinicians from different fields), this was classified as ‘multidisciplinary pain management’. Otherwise, these interventions were classified under 'treatment combinations'.”

In our experience thus far in classifying INTs, we did not have major challenges separating multidisciplinary pain management from INTs that have multiple components.

Reviewer #2: unclear definition and ambiguity of nodes/mixed interventions. For example, which is the difference between multidisciplinary node and a node with primary and secondary component? Then, which is the difference between Physical therapy and multidisciplinary node?

Nodes components are arbitrary chosen and can be subjective to a posteriori changes driven by evidence (primary and secondary components): personnel subjective judgment are based on what? time for dose, sessions, scheduling or duration of whole intervention?

Author response: Please see the detailed information we provide in the Supplemental Data A, especially the section "Definition of primary and secondary INT components". That section is our a priori definition, agreed amongst the team in the early stages of the project, of how we approach the independent assessment by at least two skilled reviewers on how the INTs were classified.

For your ease of reference, we paste here the definitions of multidisciplinary versus physical therapy

Multidisciplinary (multidisciplinary pain management) (mul)

Multidisciplinary pain management incorporates a number of intervention types, such as education (e.g., mechanisms of chronic pain, anatomy), goal setting, exercise, stress management, relaxation and imagery, meditation and aspects of psychological therapies, medication management, family member participation implemented as one package of treatment [104–106]. These may be done as individual sessions or as group sessions. If a study labelled its intervention as multidisciplinary pain management, then this was considered multidisciplinary pain management. Other studies may have combined individual interventions into a multidisciplinary program but did not specifically label it as

multidisciplinary pain management. In this case, if the reviewers agreed that the intervention included a minimum of education, exercise, psychological therapies delivered by a multidisciplinary clinician team (at least 2 clinicians from different fields), this was classified as ‘multidisciplinary pain management’. Otherwise, these interventions were classified under 'treatment combinations' (below).

Physical therapy (otherwise not falling into specific treatment combination) (pio)

Into this group fall any interventions that are generic ‘physiotherapy’ or ’physical therapy’ treatments, often at the discretion of the clinician, but otherwise not detailed or defined.

We argue this is a well explained, robust, transparent and detailed approach to handling the INTs.

Reviewer #2: the percentage of threshold used (20%?) to choose the primary or secondary components don’t reflect the reporting of primary studies. Few or none of the RCTs declare the percentage of performing an intervention. This undermine the internal and external validity of treatment nodes.

Author response: In our work so far, the expert reviewers/extractors have been able to apply our criteria well with discussion amongst the wider team as needed for conflicts / differing viewpoints.

We believe our a-priori defined approach is transparent, well-justified and clear. Yes, some authors may take a different approach, and they are welcome to do so.

Based on the above considerations we recommend:

Reviewer #2: In the main analysis to perform the subgroup analysis (by population) analysis, not only in case of inconsistency; Please, add this in the methods.

Author response: We responded to this in our prior response. This is handled as part of pre-planned sub-group analyses: “we will perform subgroup analyses ... by type of low back disorder (e.g., non-specific chronic low back pain, radicular syndrome)”

Reviewer #2: in the main analysis to perform the sensitivity analysis without the multicomponent interventions (i.e., arms with 80% of manual therapy and 20% of exercise), not only in case of inconsistency (or cited as limitation in the discussion section) in order to run only single interventions (e.g., 100% manual therapy); Please, add this in the methods.

Author response: please see our response above regarding a prior paper which in our view incorrectly excluded many studies in their main analysis, potentially leading to biased effects estimates.

We believe this is best handled as a sensitivity analysis of the wider dataset.

We will add a sensitivity analysis on this point: “Secondary treatment components (see Supplemental Data A): the impact on effect estimates of when secondary treatments are included will be assessed via a sensitivity analysis excluding those interventions with a secondary treatment component.”

Thank-you for the suggestion.

Reviewer #2: FANS, paracetamol, opioids, muscle relaxants drugs, anticonvulsants etc.. should be presented in separate nodes;

Author response: please see our response above regarding the publication of our dataset and our willingness to collaborate with researchers who are interested in modelling the data differently.

There may be different equally justified construction of INT nodes.

Reviewer #2: be careful in the assessment of imputation of medians. “If a median is available occasionally can be used directly in meta-analyses…Wan and colleagues proposed a formula for imputing a missing mean value based on the lower quartile, median and upper quartile summary statistics (Wan et al 2014) ..assuming normally distributed outcomes” https://training.cochrane.org/handbook/current/chapter-06#section-6-5-2-9 How did the authors plan to verify the normal distribution? In order to be more conservative we strongly recommend to avoid the use of medians if normal distribution cannot be proved, this is more true when authors expect to include over 700 trials. In fact, in our recent NMA on acute and subacute low back pain we found that the use of medians or unusable outcome data was present in around 40% trials https://pubmed.ncbi.nlm.nih.gov/33849907. Imputation of one/two studies in a pairwise meta-analysis can be plausible (occasionally) but what about the imputation of medians in 40% (more or less) of 700 studies?

Author response: In our current extracted data set we found that only around 3% of the studies were imputed due to a median. As the Cochrane collaboration writes: ”However, imputation may be reasonable for a small proportion of studies comprising a small proportion of the data if it enables them to be combined with other studies for which full data are available. Sensitivity analyses should be used to assess the impact of changing the assumptions made.” (https://training.cochrane.org/handbook/current/chapter-06#section-6-5-2-7) Hence we will perform a sensitivity analysis by excluding studies with imputed values as stated in our protocol paper.

This was included in the sensitivity analyses in the previously revised submission in the line "• Excluding studies with imputed missing standard deviations and imputed medians."

VERSION 3 – REVIEW

REVIEWER	Silvia Gianola I.R.C.C.S. Orthopedic Institute Galeazzi, Clinical Epidemiology Unit
REVIEW RETURNED	11-Nov-2021

GENERAL COMMENTS	Thank you for the opportunity to review this registered protocol of a systematic review with network meta-analysis (NMA). However, we felt that some responses are inappropriate. A peer review is an effort of resources and time to improve scientific quality of research between peers, thus we believe that this role should be respected. A NMA is a difficult study design, questions are made to better understand and present methods to provide a clear reporting of results and not for getting into troubles the group of authors. Peer reviewers are not interested on the “use yours raw data to conduct re-analysis using alternative methods” but to improve the scientific quality of the protocol and to avoid major issues in the subsequent publication of results. It is still unclear if the planned subgroup analysis (by population) will be done by default as primary analysis or done only in case of inconsistency (see Assessment of similarity and homogeneity assumptions, Subgroup and sensitivity analyses paragraphs). it seems that “we will perform subgroup analyses to explore whether inconsistency/heterogeneity and group differences in the outcomes are influenced by type of low back disorder (e.g., non-specific chronic low back pain, radicular syndrome), type of treatment (e.g., surgical, pharmacological) or by exclusion of the multidisciplinary node and the physical therapy (otherwise not falling into specific treatment combination) node from analyses”. Of course we agreed with the authors that there are often multiple ways to address a problem (NMA field included). Run the subgroup analysis by default as main analysis answer to majority of our questions (eligibility criteria, nodes). Anyway, I leave the decision of primary analyses along with the subgroup population to the editors. Silvia Gianola and team Greta Castellini and Silvia Bargeri